1·5 Prime Number and Composite Number

The factors of some numbers are written below :

Numbers	Factors
2	1, 2
5	1, 5
13	1, 13

Observe : The factors of each of the numbers 2, 5 and 13 in the above table are only 1 and the number itself. Such numbers are called **Prime Numbers.**

Numbers	Factors
6	1, 2, 3, 6
9	1, 3, 9
12	1, 2, 3, 4, 6, 12

Again, the factors of 6, 9 and 12 are 1 and there are numbers more than one other than that. Such numbers are called **Composite Numbers.**

1·6 Co-Prime Numbers

8 and 15 are two natural numbers.

Here, $8 = 1 \times 2 \times 2 \times 2$ and $15 = 1 \times 3 \times 5$

Observe : The factors of 8 are 1, 2, 4, 8 and the factors 15 are 1, 3, 5, 15. We can see that there is no common factors except 1 of the numbers 8 and 15. That is why numbers 8 and 15 are Co-Prime numbers to each other.

Again, there is no common factors except 1 of the numbers 10, 21 and 143. So, the numbers are co-prime to each other.

If the common factor of two or more numbers is only 1, then the numbers are Co-Prime to each other.

Activity :

1. Write ten prime numbers of 2, digits
2. Determine the prime numbers from 101 to 150.
3. Determine co-prime numbers of the following pairs :
   (a) 16, 28 (b) 27, 38 (c) 31, 43 (d) 210, 143

1·7 Divisibility

Divisible by 2

Writing some multiples of 2 we get

$2\times0 = 0, 2\times1 = 2, 2\times2 = 4, 2\times3 = 6, 2\times4 = 8,$
$2\times5 = 10, 2\times6 = 12, 2\times7 = 14, 2\times8 = 16, 2\times9 = 18$ etc.

Let us observe the process of the product. Any number when multiplied by 2, the ones place of the product will be 0, 2, 4, 6 or 8. Therefore, the numbers having 0, 2, 4, 6 or 8 in ones' place will be divisible by 2. Such numbers are called Even Number.

If the digit of the ones place of a number is 0 (zero) or an Even Number, then the number is divisible by 2.

Divisible by 4

If the number 3512 is written in terms of place value, we get,

$3512 = 3000 + 500 + 10 + 2$

Here, 10 is not divisible by 4 but the digits on the left of tens place is divisible by 4·

Again, $3512 = 3000 + 500 + 12$

Here, 12 is divisible by 4. Therefore, 3512 is divisible by 4, i.e since the number formed by the digits of ones and tens is divisible by 4, the given number will be divisible by 4.

If any number formed by the digits of ones' and tens' place of a number is divisible by 4, then the given number will be divisible by 4. Again, if the number of both tens' and ones' place is 0, then the number will also be divisible by 4.

Divisible by 5

Let us write some multiples of 5 :

$5\times0 = 0$, $5\times1 = 5$, $5\times2 = 10$, $5\times3 = 15$, $5\times4 = 20$,
$5\times5 = 25$, $5\times6 = 30$, $5\times7 = 35$, $5\times8 = 40$, $5\times9 = 45$ etc.

If we observe the process of products,we see that each of the product has 0 or 5 in its ones' place. Therefore ,if any number has 0 or 5 in its ones place, then that number will be divisible by 5.

If any number has 0 or 5 digits in its ones place, then the number will be divisible by 5.

Divisible by 3

1 4 7

place value of 7= 7

place value of 4 = 40 = 36 + 4 = (3 × 3 × 4) + 4

place value of 1 = 100 = 99 + 1 = (3×3×11) + 1

Here, 3 × 3 × 4 and 3 × 3 × 11 numbers are divisible by 3 and the sum of the digits in ones, tens and hundreds place is = 1 + 4 + 7 = 12 which is divisible by 3.

∴ The number 147 is divisible by 3.

Again, let us consider a number148:

1 4 8

place value of 8 = 8

place value of 4 = 40 = 36 + 4 = (3 × 3 × 4) + 4

place value of 1 = 100 = 99 + 1 = (3×3×11) + 1

Here, 3 × 3 × 4 and 3 × 3 × 11 numbers are divisible by 3 but the sum of digits in ones, tens and hundreds place is = 1 + 4 + 8 = 13 which is not divisible by 3.

∴ The number 148 is not divisible by 3.

If the sum of the digits of a number is divisible by 3, then the number is also divisible by 3.

Divisible by 6

Any number divisible by 2 and 3 will also be divisible by 6.

Divisible by 9

Let us consider a number 378.

3 7 8

place value of 8 = 8

place value of 7 = 70 = 63 + 7 = (7 × 9) + 7

place value of 3 = 300 = 297 + 3 = (33 × 9) + 3

Here, each of the numbers 7 × 9 and 33 × 9 is divisible by 9 and the summation of digits in ones, tens and hundreds' place is 3 + 7 + 8 = 18 which is divisible by 9. So, 378 is divisible by 9.

If the sum of the digits of any number is divisible by 9, then the given number will also be divisible by 9.

Activity :

Write numbers consisting of 3 or 4 or 5 digits which are divisible by 3 and 9.

Example 1: When Zarif asked Jawad to write six numbers consisting of one digit, he wrote 2, 0, 3, 8, 7 and 4. Writing 475□2, Zarif told Jawad there are some digits which, if put in the blank place, make numbers which are divisible by 3 each time.

a) Separate prime numbers from the numbers written by Jawad and explain why they are prime numbers.

b) Show that the subtraction of the greatest and the smallest number formed by the numbers written by Jawad is divisible by 9.

c) Determine which number will be put in the blank □.

Solution:

(a) The digits written by Jawad are : 2, 0, 3, 8, 7 and 4. Of these digits the prime numbers are 2, 3, 7

Because, 2=1 x 2, 3=1 x 3, 7=1 x 7

Therefore, the factor of 2, 3, 7 is 1 and the number itself.

(b) The digits written by Jawad are : 2, 0, 3, 8, 7 and 4.
Here, 8>7>4>3>2>0
Therefore, the greatest number formed by 2, 0, 3, 8, 7 and 4 is 874320 and the smallest number = 203478.
Now, the subtraction of the greatest number and the smallest number formed = 874320-203478 = 670842
Again, the sum of the digits of the number 670842 = 6+7+0+8+4+2 = 27, which is divisible by 9.
So the subtraction of the greatest and the smallest number formed is divisible by 9 (shown)

(c) The sum of the digits used in 475□2 is 4+7+5+2 = 18; which is divisible by 3. So, if 0 is placed in □ the number will be divisible by 3.
If 3 is added to the sum of the digits, it becomes 18+3 = 21 which is divisible by 3. Therefore, if 3 is put in the number it formed it will be divisible by 3 Similarly, 18+6 = 24 which is divisible by 3.
So, if either of 6 or 9 is placed in □, the number formed will also be divisible by 3. Therefore, if any of the digits 0, 3, 6, 9 is put in □, the number formed will be divisible by 3 in each case.

Exercise 1·2

1. Write down the prime numbers between 30 to 70 :
2. Determine which of the following pairs are co-prime :
   (a) 27, 54 (b) 63, 91 (c) 189, 210 (d) 52, 97
3. Which of the following numbers are divisible by the numbers as indicated?
   (a) 545, 6774, 8535 by 3 (b) 8542, 2184, 5274 by 4
   (c) 2184, 1074, 7832 by 6 (d) 5075, 1737, 2193 by 9
4. Which digits are to be put in the blanks so that each of the following number is divisible by 9 ?
   (a) 5 ☐ 4723 (b) 812 ☐ 74 (c) ☐ 41578 (d) 5742 ☐
5. Determine the smallest number of 5 digits which is divisible by 3.
6. Determine the greatest number of 7 digits which is divisible by 6.

7. Determine whether the greatest and the smallest numbers formed by the digits 3, 0, 5, 2, 7 are divisible by 4 and 5 or not.

1·8 Highest Common Factor (H.C.F.)

We know that factors of 12 are 1, 2, 3, 4, 6 and 12
and factors of 30 are 1, 2, 3, 5, 6, 10, 15 and 30
Here, common factors of 12 and 30 are 1, 2, 3 and 6
Among those common factors, highest common factor is 6.
∴ H.C.F. of 12 and 30 is 6.

The greatest of the factors among the common factors of the given numbers is called the Highest Common Factor of those given numbers. It is shortly written as H.C.F.

Again, we know, prime factors of 12 are 2, 2, 3 and prime factors of 30 are 2, 3, 5

∴ Common prime factors of 12 and 30 are 2, 3
∴ H.C.F. of 12 and 30 = $2 \times 3 = 6$

The product of the common prime factors of the given numbers is the H.C.F. of the given numbers.

Example 1. Determine H.C.F. of 28, 48 and 72 by the help of factors and both prime factors.

Solution : Determination of H.C.F. by the help of factors :
Here, factors of 28 are 1, 2, 4, 7, 14, 28
factors of 48 are 1, 2, 3, 4, 6, 8, 12, 16, 24, 48
and factors of 72 are 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72
∴ Among common factors of 28, 48 and 72, the highest common factor is 4.

∴ H.C.F. of 28, 48 and 72 is 4.
Determination of H.C.F. by prime factors :
Here, prime factors of 28 are 2, 2, 7
prime factors of 48 are 2, 2, 2, 2, 3
and prime factors of 72 are 2, 2, 2, 3, 3
∴ Common prime factors of 28, 48 and 72 are 2, 2
∴ H.C.F. of 28, 48 and 72 $= 2 \times 2 = 4$.

Determination of H.C.F. by division method :

Example 2. Determine the H.C.F. of 12 and 30.

Solution : Here, 12) 30 (2
24
6) 12 (2
12
0

The last divisor is 6.

∴ H.C.F. of 12 and 30 is 6.

Example 3. Determine H.C.F. of 28, 48 and 72.

Solution : 28) 48 (1
28
20) 28 (1
20
8) 20(2
16
4) 8 (2
8
0

Again, 4) 72 (18
4
32
32
0

Here, the last divisor is 4.

∴ H.C.F. of 28, 48 and 72 is 4.

Activity:
Write the smallest number consisting of 4 digits and the greatest number consisting of 3 digits in each of which 8 will be in ones' place. Find the H.C.F. of these two numbers by prime factors and division process.

1·9 Lowest Common Multiple (L.C.M.)

We know, multiples of 4 are : 4, 8, 12, 16,20, 24, 28, 32, 36, 40,44, 48 etc.
multiples of 6 are : 6, 12, 18, 24, 30, 36, 42, 48 , 54 etc.
and multiples of 8 are : 8, 16, 24, 32, 40, 48, 56, 64 etc,

We can see, common multiples of 4, 6 & 8 are 24, 48 etc and among them the lowest multiple is 24.
∴ L.C.M. of 4, 6 and 8 is 24.

The lowest common multiple of two or more numbers is called their Lowest Common Multiple (L.C.M.).

Again, if we write the prime factors of 4, 6, 8,we get

$4 = 2 \times 2,\quad 6 = 2 \times 3,\quad 8 = 2 \times 2 \times 2$

In prime factors of 4, 6, 8 the maximum repetation of 2 is 3 times and that of 3 is 1 time. So, successive multiplication of 3 times of 2 and 1 time of 3 gives $2\times2\times2\times3$ or 24 which is L.C.M. of the given numbers.

Determination of L.C.M. by Euclid's Process :

Example 4. Determine L.C.M. of 12, 18, 20, 105.

Solution :

2	12, 18, 20, 105
2	6, 9, 10, 105
3	3, 9, 5, 105
5	1, 3, 5, 35
	1, 3, 1, 7

∴ Required L.C.M. $= 2 \times 2 \times 3 \times 5 \times 3 \times 7 =$ 1260

It is to be noted from the above example:

- The numbers are written in a row placing comma (,) in betwen them and a line ∟ is drawn under them.
- At least two of the given numbers are divided by a prime number. The quotients of the numbers divisible by the factor are written below the line remaining unchanged along with indivisible numbers.
- Similar activity has been done with the number of the next row.
- The process continues until the numbers in the last row are co-prime numbers.

➢ The required L.C.M. is the successive multiplication of the divisors and the numbers of the last row.

1·10 Relationship between L.C.M. and H.C.F.

Prime factors of any two numbers 10 and 30 are determined below :

$10 = 2 \times 5$

$30 = 2 \times 3 \times 5$

$\therefore$ H.C.F. of 10 and 30 $= 2 \times 5 = 10$ and L.C.M. $= 2 \times 3 \times 5 = 30$

Again, product of 10 and 30 $= 10 \times 30 = (2 \times 5) \times (2 \times 3 \times 5)$

$= \text{H.C.F.} \times \text{L.C.M.}$

$\therefore$ Product of two numbers is equal to the product of L.C.M. and H.C.F. of those numbers.

Product of two numbers = H.C.F. of two numbers × L.C.M. of two numbers.

Activity :
Determine the H.C.F. and L.C.M. of two or three numbers consisting two digits and perform a quick quiz contest among one of another.

Example 5. Determine L.C.M. of 30, 36 and 40 by the help of prime factors.

Solution : Here, $30 = 2 \times 3 \times 5$

$\therefore$ prime factors of 30 are 2, 3, 5

$36 = 2 \times 2 \times 3 \times 3$

$\therefore$ prime factors of 36 are 2, 2, 3, 3

and $40 = 2 \times 2 \times 2 \times 5$

$\therefore$ prime factors of 40 are 2, 2, 2, 5

$\therefore$ L.C.M. of 30, 36, 40 $= 2 \times 2 \times 2 \times 3 \times 3 \times 5 = 360$

$\therefore$ Required L.C.M. is 360

Example 6. Determine the H.C.F. of 42, 48 and 56 by division method :

Solution : Here,

```
42 ) 56 ( 1
     42
     14) 42 (3
         42
          0
```

$\therefore$ Last divisor is 2

$\therefore$ Required H.C.F. is 2

Again,

```
14 ) 48 ( 3
     42
      6 ) 14 ( 2
          12
           2) 6 ( 3
              6
              0
```

Example 7. What is the greatest number which divides 365 and 463 with the remainders 5 and 7 respectively ?

Solution: With the remainders 5 and 7 respectively, when 365 and 463 are divided by the greatest number, the greatest number will be the H.C.F. of (365 – 5) or 360 and (463 – 7) or 456.

Here, 360) 456 (1
360
96)360(3
288
72) 96 (1
72
24) 72 (3
72
0

∴ H.C.F of 360 and 456 is 24.

∴ Required greatest number is 24.

Example 8. What is the greatest number which divides 57, 98 and 183 with no remainder ?

Solution : Required greatest number will be the H.C.F. of 57, 98 and 183.

Here, $57 = 3 \times 19$, $93 = 3 \times 31$ and $183 = 3 \times 61$

∴ H.C.F. of 57, 93 and 183 is 3.

∴ Required greatest number is 3.

Example 9. What is the smallest number when added by 5 the summation will be divided by 16, 24 and 32 ?

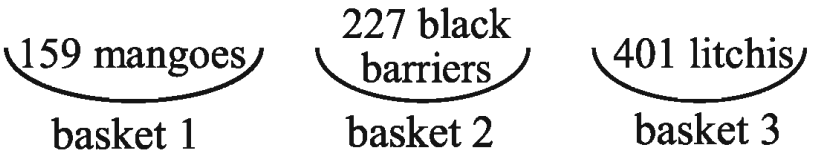
Solution : The required number which will be less than 5 of the L.C.M. of 16, 24 and 32.

2	16, 24, 32
2	8, 12, 16
2	4, 6, 8
2	2, 3, 4
	1, 3, 2

$\therefore$ L.C.M. of 16, 24 and 32 = $2 \times 2 \times 2 \times 2 \times 3 \times 2 = 96$

$\therefore$ Required least number will be (96 – 5) or 91.

Example 10 :



a) Determine the factors of 159 and then separate the prime factors.

b) If 9 mangoes, 7 black berries and 1 litchi get rotten, then find out the L.C.M. of the remaining fruits by Euclid's method.

c) What is the highest number of boys among whom the fruits can be equally distributed with 3 mangoes, 6 black berries and 11 litchis left?

Solution

(a) 159 = 1 X 159

= 3 X 53

the factors of 159 are 1, 3, 53 and 159. Of them 3 and 53 are prime factors.

(b) The number of good mangoes in basket 1 is 159-9 = 150
The number of good mangoes in baskets 2 is 227-7 = 220.
The number of good mangoes in basket 3 is 401-1 = 400.

Now,

2 | 150, 220, 400
2 | 75, 110, 200
5 | 75, 55, 100
5 | 15, 11, 20
3, 11, 4

$\therefore$ The L.C.M. of 150, 220 and 400 is 2 X 2 X 5 X 5 X 3 X 4 X 11 = 13200

(c) Here, 159-3 = 156

227-6 = 221

401-11 = 390

The number of boys will be the H.C.F. of 156, 221 and 390.

Here,

156)221(1
156
65)156(2
130
26)65(2
52
13)26(2
26
0

Again,

13) 390 (30
39
0
0
0

So, the L.C.M of 156, 221 and 390 is 13.
So, the number of boys is 13

Alternative Method

2|156
2|78
3|39
13

So, 156 = 2 x 2 x 3 x 3 x 13

13|221
17

So, 221 = 13 x 17

2|390
3|195
5|65
13

So, 390 = 2 X 3 X 5 X 13
So, the prime factor of 156, 221 and 390 is 13
So, the number of boys is 13.

Exercise 1·3

1. Determine H.C.F. using prime factors :
   (a) 144, 240, 612 (b) 525, 495, 570 (c) 2666, 9699
2. Determine H.C.F. by division method :
   (a) 105, 165 (b) 385, 286, 418
3. Determine L.C.M. using prime factors :
   (a) 15, 25, 30 (b) 22, 88, 132, 198 (c) 24, 36, 54, 72, 96
4. Determine L.C.M. by Euclid's method : :
   (a) 96, 120 (b) 35, 49, 91 (c) 33, 55, 60, 80, 90

5. What is the greatest number which divides 100 and 184 with the remainder 4 in every time ?
6. What is the greatest number which divides 27, 40 and 65 with remainders 3, 4, 5 respectively?
7. Which smallest number will have 5 as remainder in every case when divided by 8, 12, 18 an 24?
8. Which smallest number will have 15, 20, 25, 31 and 43 as remainder respectively when divided by 20, 25, 30, 36 and 48 ?
9. The length of a iron-sheet and a copper-sheet is 672 cm and 960 cm respectively. What will be the length of the highest piece of equal size cut from the two sheets ? Determine the number of pieces of sheets ?
10. What is the least number of 4 digits which is divided by 12, 15, 20 and 35 ?
11. What is the greatest number of 5 digits which when divided by 16, 24, 30 and 36, 10 will be remainder in each case ?
12. Four buses traverse a distance of 10 km, 20 km, 24 km. and 32 km respectively from a bus-stand after a definite interval of time. What is the minimum distance traversed by the buses when they meet together ?
13. The product of two numbers is 3380 and their H.C.F. is 13. Determine the L.C.M. of the two numbers.

Fractions

1·11 Common Fractions

In previous class, we have learnt about fractions.Here, we shall discuss common fractions. Common fractions are of three types. They are : Proper fractions, Improper fractions and Mixed fractions.